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Palaeomagnetism of Permian and Triassic red beds of NW Spain and implications for the tectonic evolution of the Asturian–Cantabrian Arc

J. M. Parés[1], R. Van der Voo[2] and J. Stamatakos[2]

[1] *Institute of Earth Sciences "Jaume Almera", CSIC, Marti i Franques s/n,* 08028 *Barcelona, Spain*

[2] *Department of Geological Sciences, University of Michigan, Ann Arbor,* MI 48109-1063, *USA*

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SUMMARY

A palaeomagnetic study of Permo-Triassic cover rocks in the Asturian–Cantabrian Arc has revealed characteristic magnetizations in 136 samples from 14 sites, yielding a mean palaeopole for each of the three regions studied (Cabo de Peñas, Villaviciosa, and Tudanca). Previous work on remagnetized Palaeozoic formations in the area revealed Permian (or younger) tightening of the Arc on the basis of palaeomagnetic declinations that show relative rotations of more than 100°. According to the new palaeopoles obtained in this study, these rotations are not present to any comparable extent in the Permo-Triassic deposits, allowing us to conclude that the bulk of the rotations in the Arc is Permian in age.

Key words: palaeomagnetism, Permian, tectonics, Triassic.

INTRODUCTION

Plan-view curvature of orogenic belts has been the topic of numerous palaeomagnetic studies ever since Carey (1955) first proposed the concept of 'oroclines' and oroclinal bending. Although palaeomagnetism cannot *a priori* discriminate between the possible mechanisms by which oroclines form, the technique is well established as a powerful tool to constrain the kinematics of curvature or more simply to test whether any bending (i.e. rotations) occurred. The Sicilian–Calabrian, Umbrian, Central Appalachian, Wyoming–Idaho and Ibero–Armorican arcs are examples of oroclines that have been studied palaeomagnetically (Eldredge, Bachtadse & Van der Voo 1985; Lowrie & Hirt 1986 and references therein).

The Asturian–Cantabrian Arc (Fig. 1), a foreland thrust-and-fold belt in the external part of the Variscan Belt in northwest Spain, forms the core of the Ibero–Armorican Arc. A large number of thrusts and folds can be observed, which define the form of the Arc. The relative ages and positions of synorogenic units permit the establishment of the ages of emplacement of the main thrust units. These stratigraphic data indicate that the allochtonous units of the Cantabrian zone were emplaced in Westphalian B to Stephanian ages. The Arc has been the focus of many palaeomagnetic and structural studies investigating the orocline hypothesis (Ries & Shackleton 1976; Julivert & Arboleya 1984, 1986; Matte 1986; Perroud 1986; Pérez-Estaún *et al.* 1988; Hirt *et al.* 1992). The Arc is unusual in that it is concave toward the foreland, with thrust sheets converging towards the Arc's centre. Most of the previous studies concluded that the Arc was originally not linear, but possessed a primary curvature of about half its present form. Then the Arc was tightened as deformation progressed by rotations of thrust sheets and generation of radial folds, all inferred to have taken place in the Upper Carboniferous.

In recent studies however, we have reported palaeomagnetic results from remagnetized Devonian carbonates in the Asturian–Cantabrian Arc (Parés *et al.* 1994; Stamatakos, Parés & Van der Voo 1995), which indicate that rotations occurred after initial thrusting and folding in the Permian or later. Our reasoning is based upon the following.

(1) The Devonian carbonates have a secondary magnetization carried by magnetite. Remagnetization is demonstrably younger than the folding, which took place during the Westphalian B to Stephanian stages of the Upper Carboniferous, and from a comparison with reference inclinations from the rest of Europe appears to be Permian.

(2) Re-examination of previous results of red clastic rocks and haematite-bearing carbonates indicates that these rocks are also remagnetized. Unlike the grey carbonates, the secondary directions of those red beds appear to be a syn-folding magnetization acquired during Late Carboniferous folding. Then palaeomagnetic studies as a whole show that the remagnetizations have been acquired during the Late Carboniferous–Permian interval.

(3) The overall pattern of declinations in all these rocks correlates with structural trends of the Arc, suggesting significant oroclinal bending (up to about 100°).

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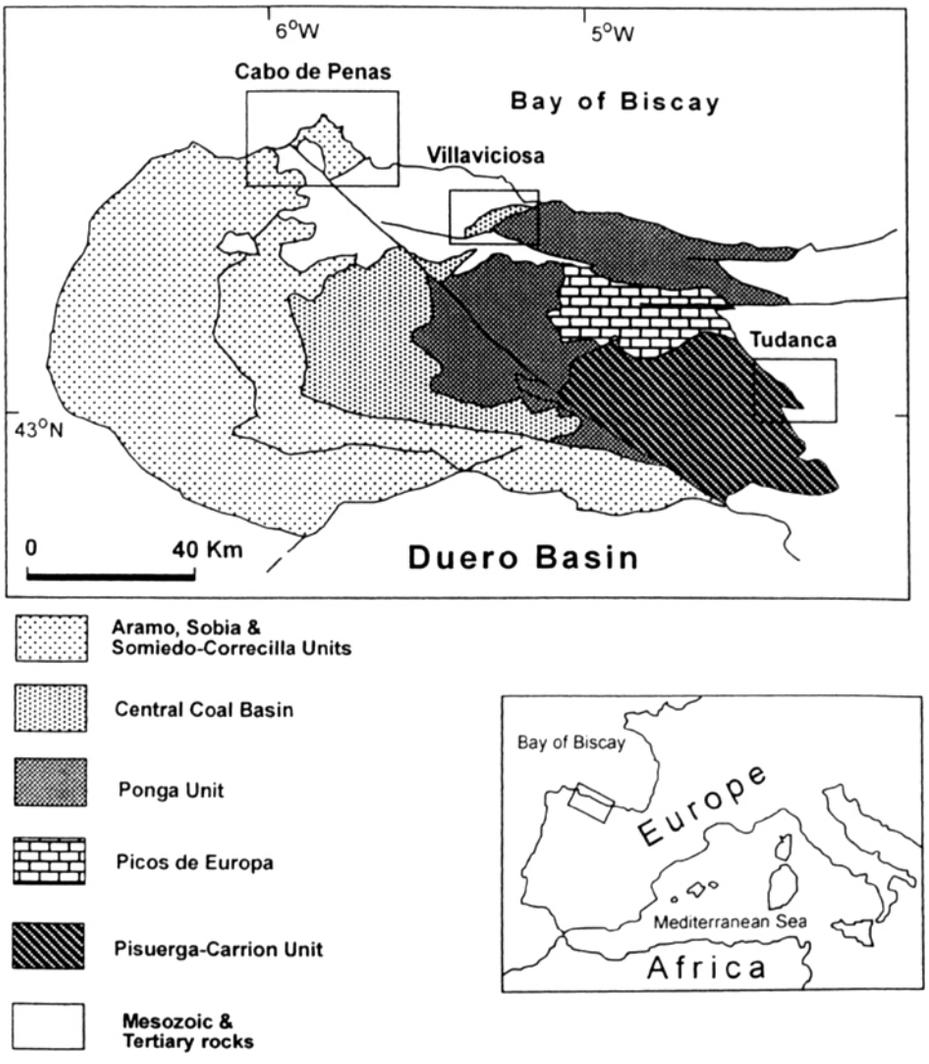


Figure 1. Sketch of the Cantabrian Arc showing the main tectonic units, including outlines of the zones studied (Fig. 2).

(4) Oroclinal rotations in the arc must have occurred during or possibly after the Permian, thus post-dating the main phase of Variscan compression.

In the northern and eastern parts of the Asturian–Cantabrian Arc, the deformed Palaeozoic rocks are unconformably covered by Permian and Triassic deposits. Thus, if rotations were younger than the Permian, they would have affected the cover rocks. Except for early work by Clegg *et al.* (1957), who interpreted a present-day field direction as Triassic, the first Permo-Triassic palaeomagnetic results were obtained by Vandenberg (1980). He reported declination discrepancies in Permo-Triassic and Late Jurassic rocks located in the northern part of the Asturian–Cantabrian Arc, and argued that the sampled rocks belonged to a crustal block that extends off-shore. This hypothetical crustal terrane, which he named the Danois Block, presumably did not rotate with Spain during the opening of the Bay of Biscay in the Cretaceous. While this hypothesis has been challenged by more recent work (Schott & Peres 1987), the observed divergent declinations could be an indication that at least some of the 100° of oroclinal rotations may have occurred after Permo-Triassic times. Intrigued by these results in the Permo-Triassic rocks and by the apparently young age of rotations recorded by the Palaeozoic magnetization, we undertook a palaeomagnetic study of the available exposures of the Permo-Triassic red beds along the northern margin of the Asturian–Cantabrian Arc, in order to document the existence and constrain the timing of these rotations better.

GEOLOGICAL SETTING AND SAMPLING

The Cantabrian Arc comprises unmetamorphosed sedimentary rocks that include a pre-orogenic succession of Precambrian rocks and three main Palaeozoic sequences (Cambro-Ordovician, Siluro-Devonian and Carboniferous). Permo-Triassic rocks were deposited on the Palaeozoic strata with a major angular unconformity: presently they are preserved only in the northern and eastern parts of the Arc. For this study we sampled the Permo-Triassic rocks in three localities (Villaviciosa, Tudanca and Cabo de Peñas; Fig. 1).

In the Villaviciosa area (Fig. 2), three main formations can be recognized in the Permian succession (Fig. 3). Unconformably on Upper Stephanian lies the Sotres Formation, a sequence of

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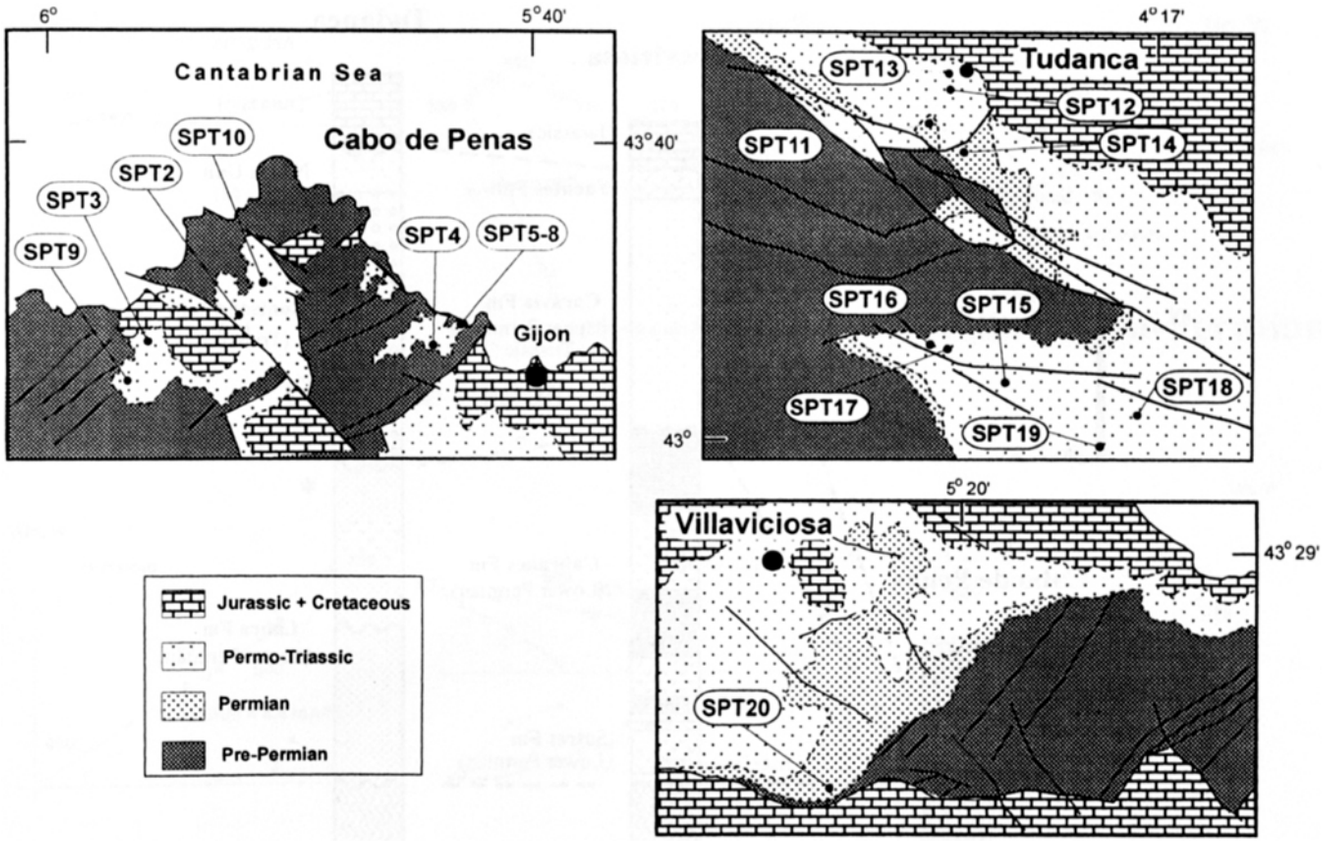


Figure 2. Geological settings of the sampling localities (Cabo de Penãs, Villaviciosa and Tudanca).

dark shales, algal limestones, volcaniclastic sediments, grey limestones, conglomerates, and lava flows. A thick volcanic and volcano-sedimentary succession with interbedded shales, sandstones and siliceous conglomerates of the Cabranes Formation (Martinez-Garcia 1991) overlies the Sotres Formation. Megafloral remains and pollen date both the Sotres and Cabranes formations as Lower Permian (Autunian) (Wagner & Martinez-Garcia 1982). Separated from the Cabranes sequence by a minor unconformity is the Caravia Formation, made up of alternating shales and sandstones, sometimes with siliceous conglomerates at the base. The age of the Caravia Formation has not yet been determined, but, because it lies between Autunian sediments and Buntsandstein (Lower Triassic) deposits in more easterly parts of the Cantabrian Province, it is considered as Saxonian (Upper Permian) (Burkhardt 1976; Gervilla *et al.* 1973; Manjon, Guitierrez Claverol, & Martinez-Garcia 1992).

In the Cabo de Peñas region (Fig. 2), the Permo-Triassic sequence is thinner and is composed of red conglomerates and quartzitic sandstones that grade into red clays near the top (Fig. 3). Julivert *et al.* (1973), Martinez-Alvarez, Guitierrez-Claverol & Alonso (1975) and Gervilla *et al.* (1973) considered this sequence a Buntsandstein facies and, thus, inferred a Triassic age.

On the other hand, Martinez-Garcia (1983) considered these sediments to be part of the Caravia Formation, on the basis of lithological correlation with the formations in Villaviciosa, implying an Upper Permian age. Near the top of the Cabo de Peñas section, micaceous siltstones and red and green marls with gypsum grade into Liassic limestones and dolomites (a possible equivalent of the Fuentes Formation of the Villaviciosa section; Dubar, Mouterde & Llopis 1963; Martinez-Garcia 1991; Suarez-Rodriguez 1988). If the age of the underlying red beds (Caravia Fm.?) is Permian, as postulated by Martinez-Garcia (1983), then the conformable contact between them and the overlying Liassic limestones is problematic. Either most of the Triassic section is extremely condensed or is missing entirely.

The Tudanca area (Fig. 2) contains a thick Permo-Triassic sequence (Maas 1974; Martinez-Garcia 1983; Garcia-Mondejar *et al.* 1986, 1988) that is reasonably well exposed. Recent work by Garcia-Mondejar *et al.* (1988) describes the tectono-stratigraphic evolution of the basin. The Permian Labra Formation (Maas 1974) is composed of agglomerates and breccias, tuffs, reworked tuffs, tuffaceous sandstones and shales (Fig. 3), which are characterized by major variations in thickness, due to sedimentation in a region of troughs and swells and post-depositional erosion during later pre-Triassic block tilting. Recent discoveries of tetrapods and flora support a Lower Permian age for the Labra Formation (Martinez-Garcia *et al.* 1994). Its lower strata rest unconformably on the folded Carboniferous rocks, and it is unconformably overlain by conglomerates of Triassic Buntsandstein facies. In addition, there is one intra-Permain unconformity in the Tudanca region (Garcia-Mondejar *et al.* 1988). The basal conglomerates of the Buntsandstein grade into coarse-grained sandstones, siltstones and shales of the Triassic Nansa Unit (Maas 1974). Jurassic carbonates follow, again with an unconformity, indicating some pre-Jurassic deformation (Garcia-Mondejar *et al.* 1986) in both the Villaviciosa and Tudanca areas.

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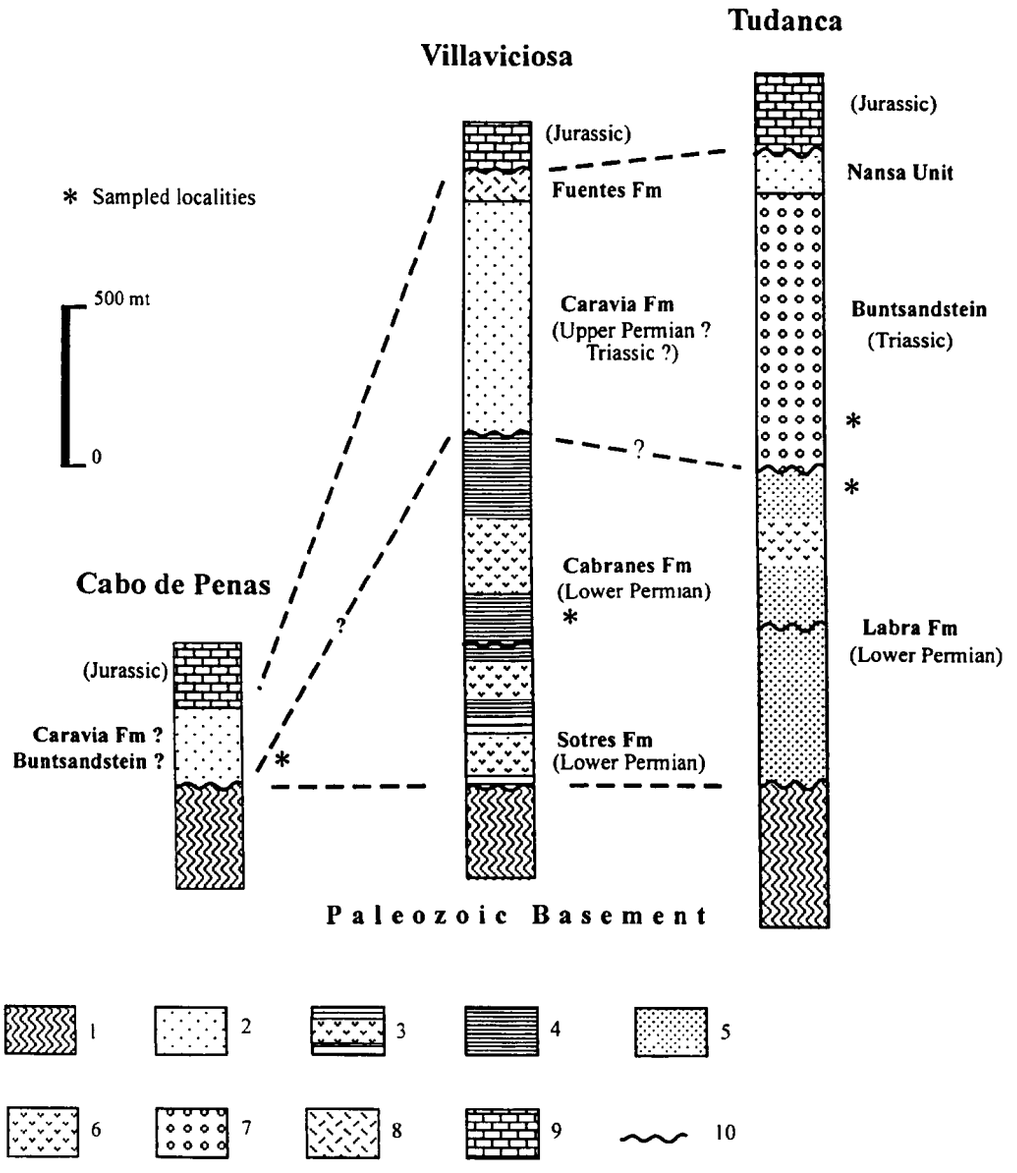


Figure 3. Schematic stratigraphic columns of the Permo-Triassic rocks in the three areas studied (Cabo de Peñas, Villaviciosa and Tudanca). 1: Palaeozoic basement, 2: shales and sandstones, 3: shales, limestones, and lava flows, 4: volcaniclastic sediments, shales and sandstones, 5: sandstones, shales and conglomerates, 6: volcaniclastic sediments and lava flows, 7: conglomerates and sandstones, 8: clays and gypsum, 9: limestones and dolomites, 10: unconformities.

PALAEOMAGNETIC RESULTS

Samples were drilled either in the field using a portable drill or from hand samples in the laboratory. In both cases, samples were oriented using a standard palaeomagnetic orientation device and magnetic compass. The magnetization is too weak to deflect the compass readings. Progressive thermal demagnetization (10 steps minimum) was performed on over 200 samples (18 sites) using an ASC thermal demagnetizer and a 3-axis cryogenic magnetometer (2G) housed in a shielded laboratory room at the University of Michigan. Magnetic components were determined by principal component analysis and visual inspection of orthogonal demagnetization diagrams. Site mean directions and associated confidence limits were calculated using Fisher statistics (Fisher 1953). Because tilting occurred about semi-horizontal axes, stepwise bedding correction tests were conducted by rotating about the line of strike, in 10 per cent increments of bedding dip.

The Permian purple siltstones from Tudanca have the highest measured values of NRM intensities, ranging from 5 to 100 mA m^{-1}. Thermal demagnetization reveals a single component of magnetization (Fig. 4), with discrete unblocking temperatures between 600° and 670 °C. Characteristic directions are either northerly or southerly, with both shallow upward and downward inclinations in *in situ* coordinates, and converge upon a NNW and shallow downward or SSE and shallow upward direction upon correction for bedding tilt (Fig. 5).

The Triassic samples from Tudanca and the samples from Villaviciosa and Cabo de Peñas yield two magnetic components: (1) a low-unblocking-temperature component parallel to the direction of the present geomagnetic field; and (2) a characteristic component with unblocking temperatures between

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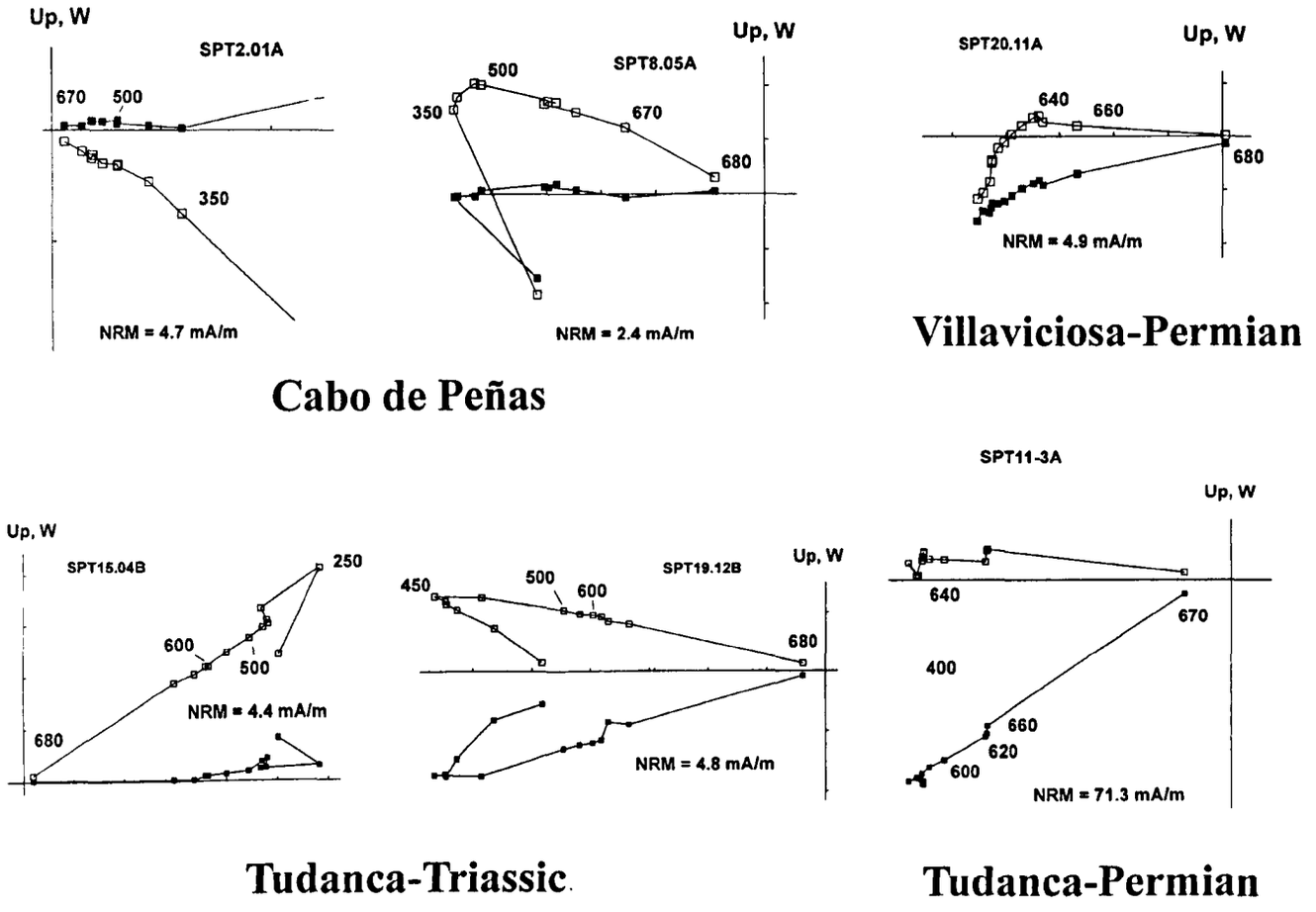


Figure 4. Representative Zijderveld (1967) diagrams of thermal demagnetization in geographic coordinates. Full (open) squares represent projections onto the horizontal (north–south vertical) plane.

640° (or lower) and 680 °C (Fig. 4). Similar to the Permian Tudanca samples, characteristic directions are either northerly and shallow downwards or antipodal southerly and shallow upwards (Fig. 5).

Characteristic directions could be obtained from 136 samples from 14 sites (Table 1). In the remaining samples, the magnetic behaviour became unstable at high temperatures, presumably due to viscous remanence acquisition in newly grown ferromagnetic minerals that masked the characteristic directions. Because of this no results could be obtained from four sites (4, 5, 9 and 18).

For the Tudanca results (sites SPT11 to SPT19), both a local and a regional fold test indicate that the magnetization is pre-folding and, given the dual-polarity magnetization, likely to be primary in age. The mean direction of the Tudanca red beds ($D/I = 340/17$) is similar to that obtained in the north by Schott & Peres (1987), who also concluded that the magnetization in the Permo-Triassic red beds was primary (Table 2).

The fold test of the Cabo de Peñas sites is not statistically significant due to the low dips of strata. However, based upon the shallow inclinations and dual-polarity magnetization, we conclude that the magnetization in this region is also primary. The overall mean direction of the Cabo de Peñas red beds at 100 per cent untilting is $D/I = 4/15$ (Table 2), which is similar in inclination but more easterly in declination than the Tudanca results.

Only one site was collected in the Villaviciosa area, given that Schott & Peres (1987) extensively sampled the Lower Permian of that region. Our mean direction does not differ significantly from theirs, so we combined the results. This combined mean is 166/-14 (given as normal polarity in Table 2).

DISCUSSION

In order to test whether the Permo-Triassic sampling areas underwent rotations, palaeomagnetic reference directions from Iberia have been used to construct an apparent polar wander path from late Palaeozoic to middle Mesozoic times. Permo-Triassic and Jurassic results from Alpine belts (the Pyrenees and Betic Cordillera) have not been included. All but one (Rio de Gallo) of the palaeopoles have a reliability Q index (Van der Voo 1993) greater than 4 (Table 3).

The combined mean palaeomagnetic pole for the Cabranes Formation in Villaviciosa falls at 52°N, 200°E ($\alpha_{95} = 7$; Table 2). This pole position is not significantly rotated from the reference polar wander path (Fig. 6). A similar conclusion is reached when the Tudanca palaeomagnetic pole (51°N, 209°E, $\alpha_{95} = 6$) is compared to the reference path, and, therefore, we conclude

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Table 1. Palaeomagnetic results from the Permo-Triassic red beds from the Cantabrian Arc grouped by location.

Site	Bedding	n/N	Dg	Ig	Dt	It	α_{95}	k
Cabo de Penas								
SPT2	65/10	14/17	7	10	9	18	7	37
SPT3	319/14	10/17	195	-18	196	-6.	7	50
SPT6	188/10	4/13	358	23	354	21	21	20
SPT7	195/32	5/17	194	-15	187	-24	14	29
SPT8	290/6	4/17	179	-8	179	-2	15	39
SPT10	354/12	9/11	175	-18	179	-18	9	34
Mean Direction	Before Tectonic correction			k = 66	a95 = 8	Dec = 5	Inc = 15	
	After Tectonic correction			k = 48	a95 = 10	Dec = 4	Inc = 15	
Tudança								
SPT11	33/25	12/12	146	11	147	-12	4	154
SPT12	300/35	13/13	325	37	340	17	7	34
SPT13	310/33	8/12	319	32	336	22	8	50
SPT14	263/23	12/12	151	-36	154	-14	5	69
SPT15	85/48	10/14	345	-25	345	22	14	13
SPT16	290/25	8/12	323	38	334	22	18	10
SPT17	110/45	7/14	353	-26	355	14	14	19
SPT19	354/10	10/12	165	-13	167	-14	12	18
Mean direction	Before Tectonic correction			k = 7	a95 = 22	Dec = 334	Inc = 12	
	After Tectonic correction			k = 72	a95 = 7	Dec = 340	Inc = 17	
Villaviciosa								
SPT20	155/21	10/10	175	-12	170	-19	11	19

n/N are the number of samples used in statistics/demagnetized; Dg (Dt) and Ig (It) are the declination and inclination of the site mean direction in geographic (stratigraphic) coordinates. α_{95} and k are the one cone of 95 per cent confidence about the mean direction and the precision parameter respectively.

Table 2. Mean directions by zones.

Zone	*Included sites*	*N*	*Dec*	*Inc*	*k*	α_{95}	*Plat*	*Plong*	*a95*
Cabo de Penas	This study	6	4	15	48	10	54	168	8
Villaviciosa	This study	1	350	19	19	11	55	200	11
	Schott and Peres (1987)	5	344	14	70	9	51	202	9
	(combined)	6	346	14	69	8	52	200	7
Tudanca	This study	8	340	17	72	7	51	209	6
	Schott and Peres (1987)	6	335	13	107	7	47	215	6

N is the number of sites included in the analysis; Dec is declination and Inc inclination; k the precision parameter and α_{95} the cone of 95 per cent confidence, Plat and Plong are the latitude (N) and longitude (E) of the virtual geomagnetic pole.

that neither the region of Villaviciosa nor of Tudanca has rotated significantly since the Late Permian–Early Triassic.

The palaeopole from the Cabo de Peñas red beds (54°N, 168°E) is located west of the other two poles from Cantabria, and thus lies on a somewhat younger part of the reference path for Iberia (Fig. 6). The difference can be explained in two ways: (1) the Cabo de Peñas section is significantly younger and reflects a Triassic rather than a Permian magnetization; or (2) the Cabo de Peñas area has been rotated clockwise during a Mesozoic or younger deformation.

The ages of the Lower Permian red bed sequences as sampled in Tudanca (sites 11 and 14) and Villaviciosa (all sites) are quite well established and their palaeomagnetic results are all of reversed polarity, consistent with the expected (Kiaman-age) polarity. The Buntsandstein facies sampled in the Tudanca area, and the sites in the Cabo de Peñas area have mixed polarity, which supports the Upper Permian or Triassic (post-Kiaman) age of the sampled rocks. Unfortunately, a more precise age assignment within the Late Permian–Triassic interval is not possible, for reasons discussed above. For the more westerly palaeopole from Cabo de Peñas, a Late Triassic age would mean that the red beds are not part of the Caravia Formation (see Fig. 3), and that the palaeopole agrees well with the Iberian polar wander path, implying no relative rotation. On the other hand, a Late Permian or Early Triassic age would mean that the Cabo de Peñas palaeopole deviates because of a clockwise rotation of about 20° (i.e. the

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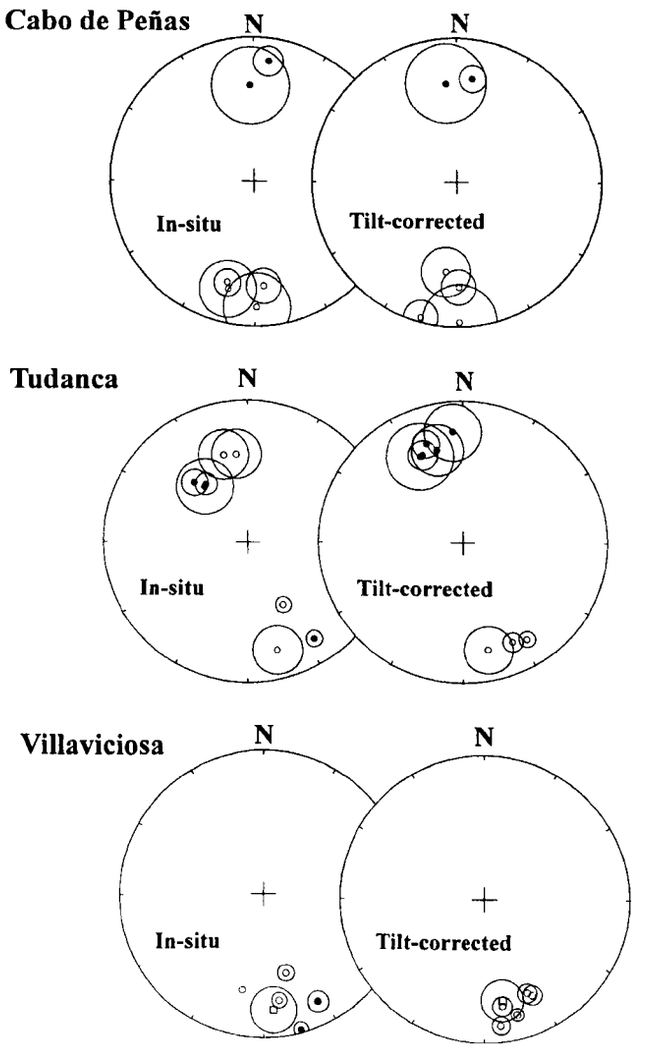


Figure 5. Site mean directions and associated cones of confidence for the Cantabrian Permo-Triassic red beds in equal-angle stereonet projections. Dots (circles) represent projections onto the lower (upper) hemisphere. On the Villaviciosa stereonet the mean directions from Schott & Peres (1987) (dots and circles) are added to our result (square).

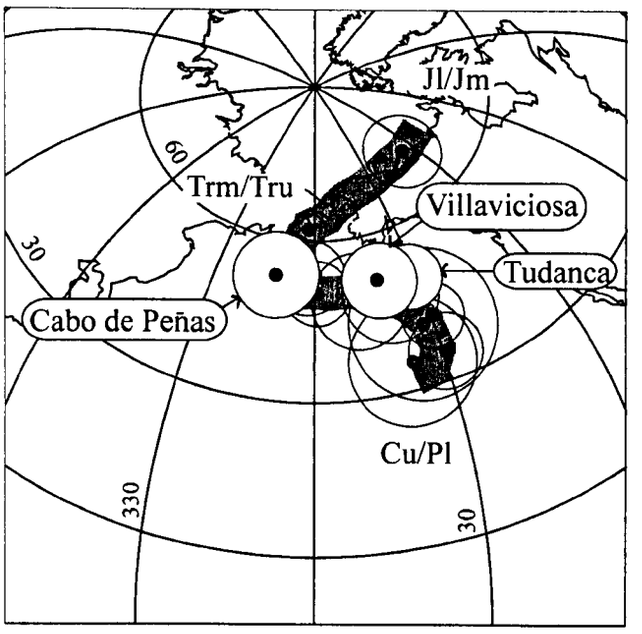


Figure 6. Late Palaeozoic to Early Mesozoic polar wander path for Iberia (based on data in Table 3) and palaeopoles for Cabo de Peñas, Villaviciosa and Tudanca (this study).

declination difference between the Triassic results from Tudanca and the results from Cabo de Peñas; see Tables 1 and 2).

Thus, if the Cabo de Peñas area has indeed rotated after deposition of the Permo-Triassic red beds, then this rotation is constrained to be no greater than 20°. This is considerably less than the rotation documented from the declination patterns of older (Palaeozoic) magnetizations in the Cabo de Peñas area (Perroud 1986; Hiet *et al.* 1992; Parés *et al.* 1994). These older magnetizations are of late Carboniferous–Early Permian age and, as mentioned in the introduction, show oroclinal bending in the Asturian–Cantabrian Arc up to about 100°. Typical declinations obtained in previous studies in the Cabo de Peñas area vary between 170° and 210°, which can be compared with the results from areas near Tudanca and Villaviciosa that yielded declination values between 130°

Table 3. Late Paleozoic–Early Mesozoic pole positions for Iberia. The palaeopoles are grouped in the groupings used for calculating the averages of Fig. 6. Abbreviations are: J = Jurassic, T = Triassic, P = Permian, C = Carboniferous, l = lower, m = middle, u = upper, k and α_{95} are the precision parameter and the cone of 95 per cent confidence about the mean pole respectively.

Rock unit, location	*Age*	*Lat*	*Long*	*k*	α_{95}	*Ref*
Viar intrus.& red beds,	Cu/Pl	43	211	110	6	Van der Voo (1969)
Bucaco red beds, N Portugal	Cu/Pl	36	211	332	7	Van der Voo (1969)
Atienza andesites, NW Iberian R.	Cu/Pl	42	208	25	14	Hernando et al. (1980)
Atienza andesites, NW Iberian R.	Cu/Pl	36	203	31	12	Van der Voo (1967)
Dikes, Catalan Coastal Range	Pl	49	202	83	9	Pares (1988)
Red beds, Rio de Gallo, Iberian R.	Pl	49	193	-	9	Turner et al. (1989)
Red beds, Catalan Coastal Range	Trm	51	181	205	6	Pares et al (1988)
Red beds, Alcazar de San Juan	Tru	63	178	47	3	Van der Voo (1967)
Red beds, Alcazar de San Juan	Tru	54	178		7	Pares and Dinares (1994)
Messejana Dike	Jl-Jm	71	236	36	7	Schott et al. (1981)

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and 165° (areas IV and V of Hirt *et al.* 1992). Thus, a best estimate of the relative rotation observed as declination differences in the Palaeozoic rocks from Cabo de Peñas and Villaviciosa/Tudanca is about 40° or twice that observed in the Permo-Triassic rocks.

Given that the Cabo de Peñas red beds are possibly of Upper Triassic age, because they show a gradual facies change with the overlying Jurassic rocks, and that they show a possible maximum rotation with respect to the other sampled areas that is much less than that observed in the Palaeozoic rocks, we conclude that the bulk of the rotations in the Cantabrian Arc are constrained to Permian time. These rotations cannot be older than the Permian, because they are recorded by post-folding magnetizations (with Permian-like inclinations) in Palaeozoic remagnetized carbonates. It now appears that these rotations (or at least a major portion of them) also cannot be younger than the Permian.

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